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Cyclic muscle contractions reinforce the acto-myosin motors and mediate the full elongation of *C. elegans* embryo

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Anna Dai, Martine Ben Amar

Laboratoire de Physique de l'Ecole normale supérieure, ENS, Université PSL, CNRS, Sorbonne Université, Université Paris Cité, F-75005 Paris, France

Abstract

The paramount importance of mechanical forces in morphogenesis and embryogenesis is widely recognized, but understanding the mechanism at the cellular and molecular level is still challenging. Due to its simple internal organization, Caenorhabditis elegans is a worthwhile study system. As demonstrated experimentally, after a first period of steady elongation due to the acto-myosin network, muscle contractions operate a quasi-periodic sequence of bending, rotation and torsion, which leads to the final 4-fold size of the embryo before hatching. How acto-myosin and muscles contribute to embryonic elongation is herein theoretically studied. A filamentary elastic model that converts stimuli generated by biochemical signals in the tissue into driving forces, explains embryonic deformation under actin bundles and muscle activities, and dictates mechanisms of late elongation based on the effects of energy conversion and dissipation. We quantify this dynamic transformation by stretches applied to a cylindrical structure mimicking the body shape in finite elasticity, obtaining good agreement and understanding for both wild-type and mutant embryos at all stages.

eLife assessment

Using the continuum theory of elastic solids, the authors suggest that periodic muscle contraction leads to elongation of C. elegans embryos by storing elastic energy that is subsequently released by extending the embryo's long axis. This **important** finding could apply to other developmental processes and be exploited in soft robotics. While the presented evidence is in principle **convincing**, features of the the theory are not explained in sufficient detail.

Introduction

Mechanical stresses play a critical role during embryogenesis, affecting complex biological tissues such as the skin, the brain, and the interior of organs. There are many studies in the last decades, for example, the pattern of the intestine (*Coulombre and Coulombre (1958)*; *Hannezo et al. (2011)*; *Li et al. (2011)*; *Ben Amar and Jia (2013)*; *Shyer et al. (2013)*), the brain cortex (*Toro and Burnod (2005)*; *Goriely et al. (2015)*; *Ben Amar and Bordner (2017)*; *Tallinen et al. (2016)*), and the cir-cumvolutions of the fingerprints (*Kücken and Newell (2005)*; *Ciarletta and Ben Amar (2012)*) that have been interpreted as the result of compressive stresses generated by growth occurring a few months post fertilization for humans. At a smaller scale, the influence of mechanical stresses superposed to cellular processes (e.g., division, migration) and tissue organization becomes more complex to be identified and quantified. For example, independently of growth, stresses may be generated by different molecular motors, among which myosin II, linked to actin filaments, is the most prevalent during epithelial cell morphogenesis (*Vicente-Manzanares et al. (2009)*) and cell motility (*Cowan and Hyman (2007)*; *Olson and Sahai (2009)*; *Palumbo et al. (2022)*). The spatial distribution and dynamics of myosin II greatly influence the morphogenetic process (***Levayer and*** Lecuit (*2012*); *Lv et al. (2022)*), as demonstrated for Drosophila (*Bertet et al. (2004)*; ***Blankenship*** et al. (*2006*); *Saxena et al. (2014)*; *Shindo and Wallingford (2014)*) and also *C. elegans* embryo (*Priess and Hirsh (1986)*; *Gally et al. (2009)*; *Ben Amar et al. (2018)*).

The embryonic elongation of *C. elegans* represents an attractive model of matter reorganization without a mass increase before hatching. It happens after the ventral enclosure and takes around 240 minutes to convert the lima-bean-shaped embryo into a final elongated worm shape: the embryo elongates along the anterior/posterior axis by four times (*McKeown et al. (1998)*). The short lifetime of the egg before hatching and its transparency makes this system an ideal system to study the forces that exist in the cortical epithelium or its vicinity. However, unlike the embryonic development of Drosophila and Zebrafish, there is neither cell migration, cell division nor a no-table change in embryonic volume (*Sulston et al. (1983)*; *Priess and Hirsh (1986)*), only a noticeable epidermal elongation drives the whole morphogenetic process of *C. elegans* in the period post en-closure. There are two experimentally identified driving forces for the elongation: the actomyosin contractility in seam cells of the epidermis, which seems to last during the whole elongation process, and the muscle activity beneath the epidermis that starts after the 1.7 ~ 1.8-fold stage. The transition is well defined since the muscle participation makes the embryo rather motile impeding any physical experiments such as laser ablation, which could be conducted and achieved in the first period (***Vuong-Brender et al. (2017a)***). Consequently, the elongation process of *C. elegans* could be divided into two stages: the early elongation and the late elongation, depending on the muscle activation, and **Fig.1(A)** shows the whole elongation process (***Vuong-Brender et al. (2017a)***). Previously, the role of the actomyosin network in the seam cells during the early elongation of *C. elegans* was investigated (*Ben Amar et al. (2018)*). Based on the geometry of a hollow cylinder composed of four parts (seam and dorso-ventral cells), a model involving the pre-stress responsible for the enclosure, the active compressive ortho-radial stress joined to the passive stress quantitatively predicts the elongation, but only up to ~ 70% of the initial length.

For the late elongation phase, the acto-myosin network and muscles drive together the full elongation during this phase (***Vuong-Brender et al. (2017a)***). Muscles play an important role since mutants with muscle defects are unable to complete the elongation process, even though the actomyosin network operates normally (*Lardennois et al. (2019)*). **Fig.1(B)** shows the schematic image of *C. elegans* body (*Zhang et al. (2011)*) with four rows of muscles, two of which are underneath the dorsal epidermis and the other two are under the ventral epidermis. As observed *in vivo, C. elegans* exhibits systematic rotations accompanying each contraction (*Yang (2017)*; *Yang Xinyi et al. (2023)*), and deformations such as bending and twisting. Eventually, the *C. elegans* embryo will complete an elongation from 1.8-fold to 4-fold. We can imagine that

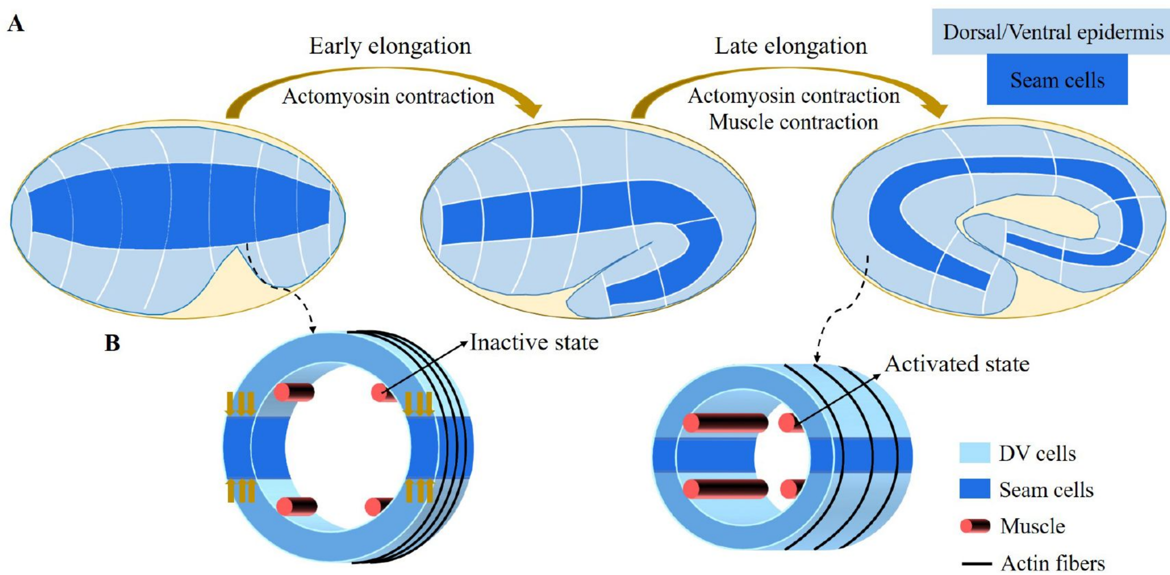


Figure 1.

(A) Overview of *C. elegans* embryonic elongation. Three epidermal cell types are found around the circumference: dorsal, ventral and lateral. (B) Schemes showing a *C. elegans* cross-section of the embryo. Small yellow arrows in the left image indicate the contraction force that occurred in the seam cell. Four muscle bands underneath the epidermis and actin bundles surround the outer epidermis.

once the muscle is activated on one side, it can only contract, and then the contraction forces will be transmitted to the epidermis on this side. So one can wonder how an elongation can occur since striated muscles can only perform cycles of contraction and relaxation, their action will tend to reduce the length of the embryo. Therefore, it is necessary to understand how the embryo elongates during each contraction and how the muscle contractions couple to the acto-myosin activity. This work aims to answer this paradox within the framework of finite elasticity without invoking cell plasticity and stochasticity which cannot be considered driving forces. In addition, several important issues at this stage remain unsettled. First of all, we may also observe a torsion of the embryo but if the muscle activity suggests a bending, it cannot explain alone a substantial torsion. Secondly, a small deviation of the muscle axis (***Moerman and Williams (2006***)) is responsible for a series of rotations, how to relate these rotations to the muscle activation (***Yang (2017***)). Since any measurement on a motile embryo at this scale is difficult, it is meaningful to explore the mechanism of late elongation theoretically. Furthermore, muscle contraction is crucial in both biological development and activities and has been studied extensively by researchers (***Tan and De Vita (2015***); ***De Vita et al. (2017***)), but how it works at small scales remains a challenge.

Using a finite elasticity model and assuming the embryonic body shape is cylindrical, we can evaluate the geometric bending deformation and the energy released during each muscular contraction on one side since after each contraction, the muscles relax and then muscles on the opposite side undergo a new contraction. This cyclic process leads to a tiny elongation of the cylinder along its symmetry axis at each contraction. Each of them is correlated with the rotational movements of the embryo (***Yang (2017***)). By repeating these pairs of contractions more than two hundred times, a cumulative extension is achieved, but it must be reduced by friction mechanisms, also evaluated by the model. Furthermore, the mechanical model explains the existence of a torque operating at the position of the head or tail by the coupling of muscle contraction with the orthoradial acto-myosin forces. Finally, the small deviation between the muscles and the central axis experimentally detected (***Moerman and Williams (2006***)) induces cyclic rotations and possibly torsion leading to fluid viscous flow inside the egg. Quantifying all these processes allows evaluating the physical quantities of the embryo such as the shear modulus of each component, the osmolarity of the interstitial fluid and the active forces exerted by the acto-myosin network and the muscles which are sparsely known in the embryonic stages.

Our mechanical model accounts for the dynamical deformations induced by the internal stimuli of a layered soft cylinder, enabling us to make accurate quantitative predictions about the active networks of the embryo. Furthermore, our model takes into consideration the dissipation that occurs in the late period just before the egg hatch. Our results are consistent with observations of acto-myosin and muscle activity (***Vuong-Brender et al. (2017a***); ***Ben Amar et al. (2018***); ***Lardennois et al. (2019***)). The architecture of our paper is illustrated in **Fig.2**.

Results

It is widely recognized that *C. elegans* is a well-established model organism in the field of developmental biology. However, it is less widely known that its internal striated muscles share similarities with skeletal muscles found in vertebrates in terms of both function and structure (***Lesanpezeshki et al. (2021***)). Being contractile, the role of the four axial muscles (as shown in **Fig.1**) in the final shaping of the embryo is nearly contra-intuitive. In this section, we aim to highlight the action of these muscles when coupled to the acto-myosin network, using a purely mechanical approach. To achieve a quantitative understanding, our method requires a detailed description of the deformation geometry of the embryo, with the body shape represented by a fully heterogeneous cylinder.

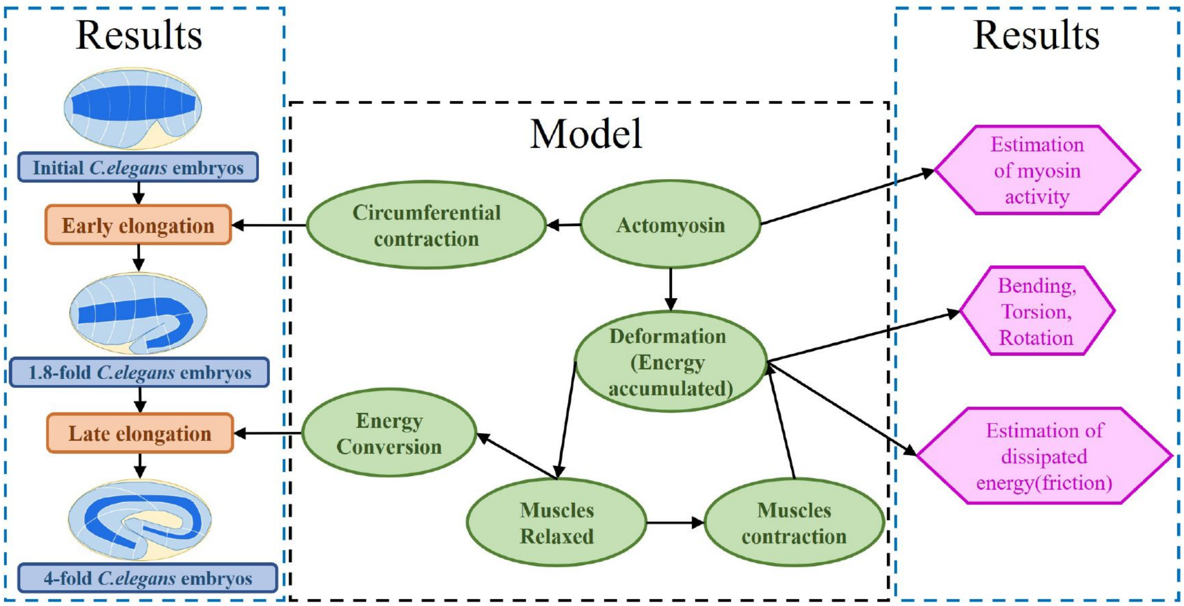


Figure 2.

Architecture of the program. The program reflects the framework of research. On the one hand, the proposed model explains the early and late elongation of the *C. elegans*, on the other hand, the early myosin activity is estimated, the deformations (bending, twisting, rotation) happening in the late period is recovered, and the estimation of energy dissipated during muscle activity is achieved.

Geometric and strain deformations of the embryo

The early elongation of *C. elegans* embryo was previously studied, which is characterized by an inner cylinder surrounded by epithelial cells located in the cortical position (***Ben Amar et al. (2018***)). The cortex is composed of three distinct cells - the seam, dorsal, and ventral cells - which exhibit unique cytoskeleton organization and actin network configurations. Among these, only the seam cells possess active myosin motors that function in the ortho-radial direction, allowing for the contraction of the circumference and thereby triggering the early elongation process. In this study, we simplify the cylindrical geometry of the body and treat the epidermis as a unified whole with effective activity localized along the circumference, supported by four muscles distributed beneath the epidermis. Muscles play no role in the early elongation stage so were not considered previously (***Vuong-Brender et al. (2017a***); ***Ben Amar et al. (2018***)). After this period of approximately a 1.8-fold increase in length, muscles parallel to the main axis and actin bundles organized along the circumference will collaborate to facilitate further elongation. **Table. 1** provides the size parameters of *C. elegans* which will be introduced hereafter.

We focus on the overall deformation of a full cylinder or a thin rod, having a length L greater than the radius R and a central vertical axis along the Z direction. Contrary to previous works (***Ciarletta et al. (2009***); ***Ben Amar et al. (2018***)), here we decide to simplify the geometrical aspect because of the mechanical complexity. The biological activity induced by the acto-myosin network and muscles is represented by active strains, and the global shape is the result of the coupling between elastic and active strains, modulated by dissipation. Active strains are generated through non-mechanical processes (e.g., biochemical processes): myosin motors and striated muscles receive their energy from ATP hydrolysis, which is converted to mechanical contractions of fibers. Due to the significant deformation observed, the central line is distorted and becomes a curve in three-dimensional space, represented by a vector $\mathbf{r}$ ($\mathbf{Z}$), as depicted in **Fig.3(A)**. Along this curve, perpendicular planar sections of the embryo can be defined, and the deformation in each section can be quantified since the circular geometry is lost with the contractions (***Kaczmarski et al. (2022***)). The geometric mapping is as follows:

$$\chi(\mathbf{X}) = \mathbf{r}(Z) + \sum_{i=1}^{3} \varepsilon a_i (\varepsilon R, \Theta, Z) \mathbf{d}_i(Z), \tag{1}$$

where the a_i represents the deformation in each direction of the section with a_i (0, 0, Z) = 0 so that the Z-axis maps to the centerline $\mathbf{r}$ (Z). The small quantity ε is the ratio between the radius R and the length L of the cylinder. Referring to the model proposed by B. Kaczmarki *et al.* (***Kaczmarski et al. (2022***)), we define the initial configuration $\mathscr{B}_0$ with material points (X, Y, Z), and the mapping function χ ($\mathbf{X}$) links the initial configuration $\mathscr{B}_0$ to the current configuration $\mathscr{B}_0$. The geometric deformation gradient is $\mathbf{F} = \text{Grad}\chi = \mathbf{F_e G}$ (***Nardinocchi and Teresi (2007***)), where $\mathbf{G}$ represents the active strain generated by the actomyosin or the muscles, and $\mathbf{F_e}$ is the elastic strain tensor. Because of the two stages through which the *C. elegans* elongates, we need to evaluate the influence of the *C. elegans* actin network during the early elongation before studying the deformation at the late stage. So, the deformation gradient can be decomposed into: $\mathbf{F} = \mathbf{F}_e \mathbf{G}_1 \mathbf{G}_0$ (***Goriely and Ben*** Amar (***2007***)) where $\mathbf{G}_0$ refers to the pre-strain of the early period and $\mathbf{G}_1$ is the muscle-actomyos in supplementary active strain in the late period. Actin is distributed in a circular pattern in the outer epidermis, see **Fig.3(B)**, so the finite strain $\mathbf{G}_0$ is defined as $\mathbf{G}_0 = Diag\,(1, g_0(t), 1)$, where $0 < g_0 < 1$ is the time-dependent decreasing eigenvalue, operating in the actin zone and is equal to unity in the other parts: $g_0(0) = 1$. In the case without pre-strain, $\mathbf{G}_0 = \mathbf{I}$. The deformation gradient follows the description of **Eq.(1)** :$\mathbf{F} = F_{ij}\, \mathbf{d}_i \otimes \mathbf{e}_j$, $i \in \{1, 2, 3\}$ and $j \in \{R, \Theta, Z\}$.

When considering a filamentary structure with different fiber directions $\mathbf{m}$, these directions are specified by two angles α and β, as outlined in (***Holzapfel (2002***)): $\mathbf{m} = \sin\alpha \sin\beta\, \mathbf{e_R} + \sin\alpha \cos\beta\, \mathbf{e_\Theta} + \cos\alpha\, \mathbf{e_Z}$, $\alpha, \beta \in [-\pi/2, \pi/2]$, where α and β lie in the range $[-\pi/2, \pi/2]$. For muscle fibers, $\alpha_m = 0$ and $\beta_m = 0$, whereas for hoop fibers in the actin network, $\alpha_a = \pi/2$ and $\beta_a = 0$. When the muscles

	Initial	1.8-fold
Radius	11.1 μm	8.2 μm
Length	50 μm	90 μm

Table 1.

Adopted real size parameters of the *C. elegans* (*Vuong-Brender et al. (2017a*); *Ben Amar et al. (2018*)).

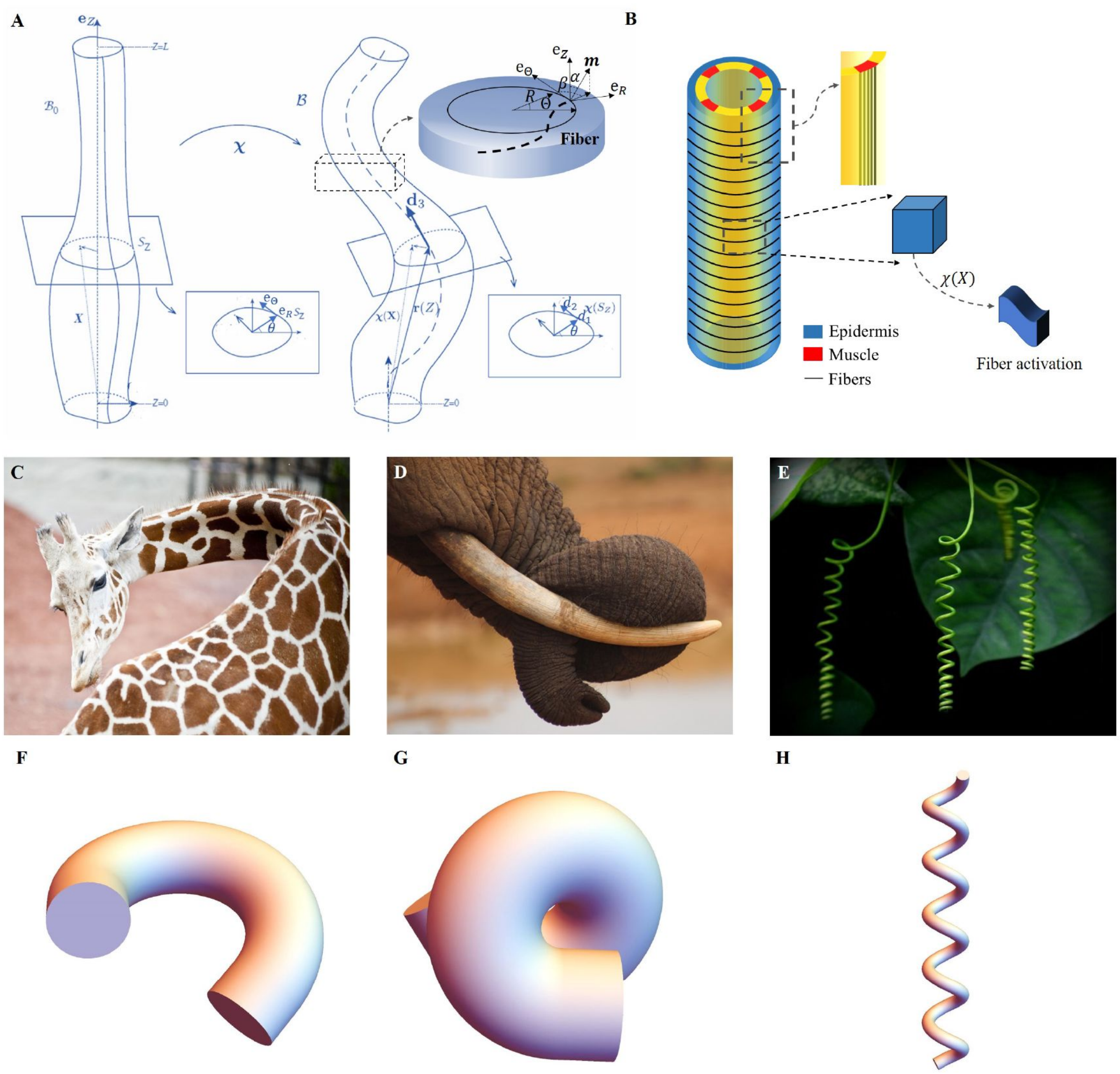


Figure 3.

(A) Cylindrical structure in the reference configuration (left) with a vertical central axis and its deformation in the current configuration (right). The deformed configuration is fully parameterized by the centerline **r** (Z) (resulting from the distortion of the central axis) and the deformation of each cross-section. (B) Schematic representation of the body shape of the *C. elegans* embryo with the cortical epidermis and the four muscles. The fibers are embedded in the cortex. The blue part representing the epidermis shows the outer distribution of the actin organized into horizontal hoop bundles when the muscles are not activated. The yellow part includes the vertical red muscles, represented by axial fibers. (C) Bending of a Giraffe neck. (D) Torsion of an Elephant trunk. (E) Plant vine twisting. (F) to (H) Deformation configuration under different activations obtained by our simulations for bending and torsion of large rods, twisting and torsion of thin rods.

are activated and bend the embryo, the actin fibers become inclined with a slope α, such that $-\pi/2 < \alpha_a < \pi/2$, $\alpha_a \neq 0$, and $\beta_a = 0$. Each active strain is represented by a tensor $\mathbf{G_i} = \mathbf{I} + \varepsilon \mathbf{g_i} \mathbf{m_i} \otimes \mathbf{m_i}$, where g_i represents the activity (g_m for the muscles and g_a for acto-myosin), and since both are contractile, their incremental activities are negative. Additional calculation details are provided in Section Methods and Materials.

To relate the deformations to the active forces induced by muscles and the acto-myosin network, we assume that the embryo can be represented by the simplest nonlinear hyperelastic model, called Neo-Hookean, with a strain-energy density given by $W(\mathbf{F}_e) = \frac{\mu}{2}\left(tr\left(\mathbf{F}_e \mathbf{F}_e^{T}\right) - 3\right)$. In cylindrical coordinates, the total energy of the system and the auxiliary energy density is:

$$\begin{aligned} W &= \varepsilon^2 \int_0^L \mathrm{d}Z \int_S V(\mathbf{F}_e, \mathbf{G}) R \,\mathrm{dR}\,\mathrm{d}\Theta, \\ V &= W(\mathbf{F_e}) - p(J-1). \end{aligned} \tag{2}$$

where $J = \det \mathbf{F}_e$. p is a Lagrange multiplier which ensures the incompressibility of the sample, a physical property assumed in living matter. When the cylinder involves several layers with different shear modulus μ and different active strains, the integral over S covers each layer. To minimize the energy over each section for a given active force, we take advantage of the weakness of ε and expand the inner variables a_i, p_i, and the potential V to obtain the auxiliary strain-energy density. Given that the Euler-Lagrange equations and the boundary conditions are satisfied at each order, we can obtain solutions for the elastic strains at zero order $\mathbf{a^{(0)}}$ and at first order $\mathbf{a^{(1)}}$. Finally, these solutions which will represent a combination of bending and torsion deformation will last the time of a muscle contraction while the acto-myosin will continue its contractile activity. The evaluation of the elastic energy under acto-myosin-muscle activity at order ε^4 can be compared with the equivalent energy of an extensible elastic rod and each basic deformation will be identified (***Kirchhoff*** and Hensel (***1883***); ***Mielke (1988***); ***Mora and Müller (2002***, 2004); ***Moulton et al. (2020a***)). The typical quantities of interest are the curvature reached during a bending event or a torsion as well as the overall elongation ζ. In the *C. elegans* embryonic system, these quantities result from the competition between the active strains due to muscles and the myosin motors and the elasticity of the body. Similar active matter can be found in biological systems, from animals to plants as illustrated in **Fig.3(C)-(E)**, they have a structure that generates internal stress/strain when growing or activity. Combining anatomy and measurement techniques, we can transform the mechanics of the body under study into a soft sample submitted to localized internal active stresses or localized internal active strains and then deduce its overall deformation mechanisms, some examples are presented in **Fig.3(F)-(H)**.

The early elongation induced by the acto-myosin

Experimental measurements during the early elongation stage reveal the embryonic diameter and the active or passive stresses, detected by fracture ablation in the body vary with the elongation fold (***Vuong-Brender et al. (2017a***); ***Ben Amar et al. (2018***)). While previous studies have extensively examined this initial stage, we have chosen to revisit it within the context of our geometry, aiming to attain complete control over our structural modeling, encompassing the geometry, shear modulus, and activity of the acto-myosin prior to muscle contraction. At the outset of the second stage, owing to the early elongation, the embryo experiences a pre-strain, which we describe in our model as $\mathbf{G}_0$. The geometry and mechanical information are depicted in **Fig.4(A)**.

Here, we need to first accurately determine $\mathbf{G}_0$ by analyzing the experimental data (***Vuong-Brender et al. (2017a***); ***Ben Amar et al. (2018***)). The cylinder can be divided into three distinct sections: the outer layer is the actin cortex, the thin ring where actin bundles concentrate and work, located within a radius range between R_2' and R_3 (as shown in **Fig.4(A)**); the middle layer ($R_2 < R < R_2'$), which is the epidermis but without actin; and the inner part ($0 < R < R_2$), where the muscle is located, along with some internal organs, tissues, and fluids. So, we treat the outer and

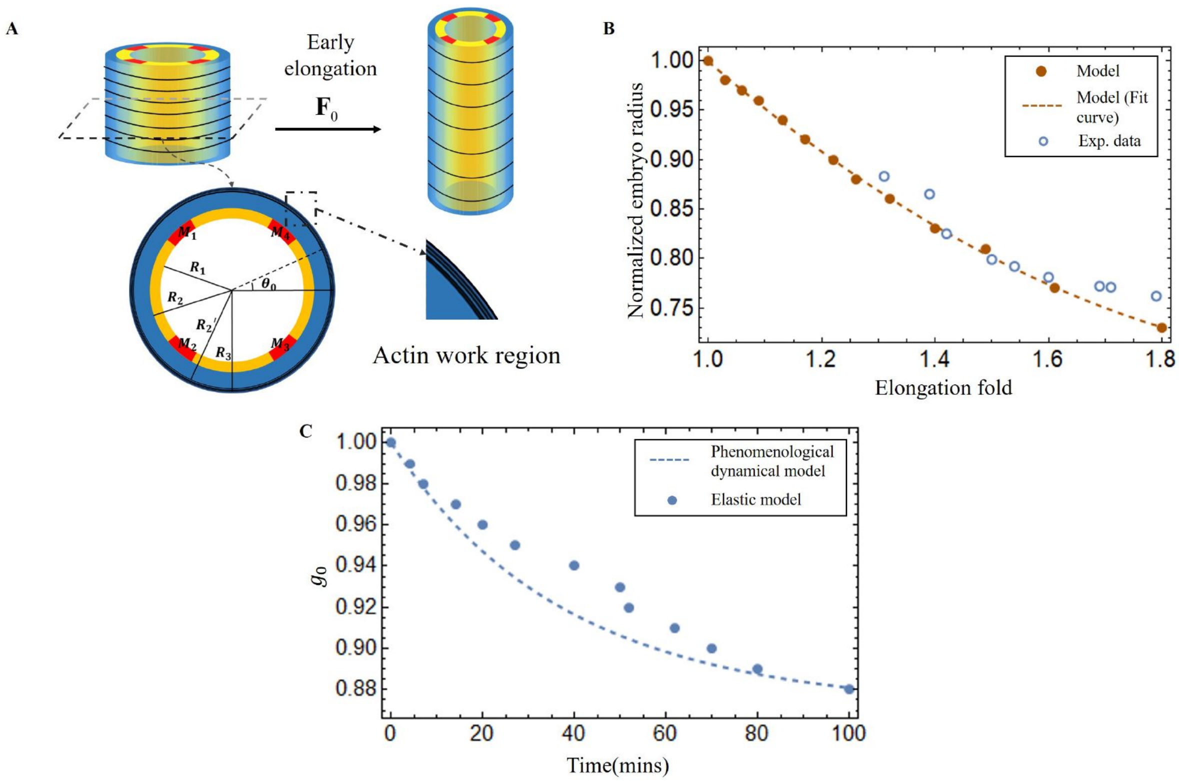


Figure 4.

(A) The schematic of early elongation and the cross-section of *C. elegans*. In the cross-section, the black circled part is the actin region ($R_2' < R < R_3$, with shear modulus μ_a), and the blue part is the epidermis layer ($R_2 < R < R_2'$, shear modulus μ_e). The central or inner part ($0 < R < R_2$, shear modulus μ_i) except the muscles which are stiffer. (B) Predictions of normalized embryo radius evolution during early elongation by the pre-strain model compared with experimental data from (***Vuong-Brender et al. (2017a)***). For the model, please refer the **Eq.(20)** in **Appendix 2**. (C) Blue dots: extraction of the parameter $g_0(t)$ from **Eq.(30)** and **Eq.(32)** in the **Appendix 2**. Blue dash line, refer to the **Eq.(3)**.

middle layers as incompressible, but the inner part as a compressible material, except the muscles. The initial deformation gradient: $\mathbf{F}_0$ =Diag($r'(R)$, $r(R)/R$, λ), and $\mathbf{G}_0$ = Diag(1, g_0, 1) with 0 < g_0 < 1 in the actin layer but with g_0 = 1 in the part without actin. $\mathbf{G}_0$ represents the circumferential strain exerted by actin during the early elongation and is a slowly varying function of time. $r(R)$ is the radius after early elongation. By applying the principles of radius continuity, radial stress continuity, and incorporating the zero traction condition on the face of the cylinder, we can determine that g_0 = 0.88 when the elongation λ = 1.8, i.e., at the completion of the early elongation. As illustrated in **Fig.4(B)**, the results of our model and the experimental data are in good agreement, this demonstrates the consistency of the geometric and elastic modeling together with the choice of a pre-strain represented by $\mathbf{G}_0$ which gives a good prediction of the early elongation. More details are displayed in **Appendix 3**.

From the first stage elongation represented by the blue dots and the blue dashed line in **Fig.4(C)** we can extract the time evolution of the contractile pre-strain $g_0(t_i)$ derived from our elastic model. To explain quantitatively $g_0(t)$, we suggest a phenomenological dynamical approach for the population of active myosin motors. This equation considers the competition between the recruitment of new myosin proteins from the epidermis cytoskeleton necessary to extend the embryo and the debonding of these myosins from the actin cables, which is damped by the compressive radial stress. It reads:

$$\frac{dX_g(t)}{dt} = \left(p_1 - p_2 X_g(t) e^{-p_3 X_g(t)}\right) \frac{t_v}{t_p} \tag{3}$$

with $X_g = 1 - g_0(t)$, p_1 is the ratio between the free available myosin population and the attached ones divided by the time of recruitment (given in minutes), while p_2 is the inverse of the debonding time of the myosin motors from the cable: $p_2 = 6\ \text{min}^{-1}$. The debonding time increases (or decreases) when the actin cable is in radial compressive (or tensile) stress, see **Appendix 3**, **Eq.(35)**. τ_v is the visco-elastic time estimated from laser ablation fracture operated in the epidermis (***Vuong-Brender et al. (2017b)***): $t_v = 6s$ and τ_p is the time required for the activation of the myosin motors $t_p = 1200s$ (***Howard and Clark (2002***)). This equation is similar to the model derived by Serra *et al* (***Serra et al. (2021***)) for the viscous stress occurring in gastrulation.

Notice that only p_1 and p_3 must be obtained by comparison of $g_0(t)$ deduced from Eq.(3) with the values deduced from our elastic model. The result of Eq.(3) with $p_1 = 0.6$ and $p_3 = 0.75$ are shown in **Fig.4(C)** with a rather good agreement.

Shape of the embryo under muscles and acto-myosin contraction

The regulation of the muscle contraction in *C. elegans* detected experimentally (***Yang (2017***)) indicates a cyclic process, where two muscles contract on one side of the embryo quite simultaneously and then stop, while on the opposite side, the two muscles start their contractions, but only after a delay. Let us consider first that only muscles are active (see the schematic **Fig.5(A)** for the structure, then (B) and (C) for the bending). In this case, due to the geometry, only a bending deformation occurs on the left for active muscles localized on left and then on the right for the symmetric muscles on right.

To be more quantitative, we assume that the left side muscles are activated during a short period with an active constant strain value g_m in the region M_1 and M_2, as shown in **Fig.5(A)-(B)**; if the muscles are perfectly vertical, $\alpha_m = \beta_m = 0$ in the initial configuration. In fact, the two muscles on the same side are not always fully in phase and one may present of small delay. For simplification, we assume them are perfectly synchronous. During the full initial period where muscles are not activated, the actin fibers are distributed in a horizontal loop on the outer surface of the epidermis, but once the muscle starts to contract, the acto-myosin network will be re-orientated (***Lardennois et al. (2019***)). The fibers will be then distributed in a sloping pattern causing eventually the twisting of the embryo, see the schematic diagram shown in **Fig.5(D)-(F)**. When

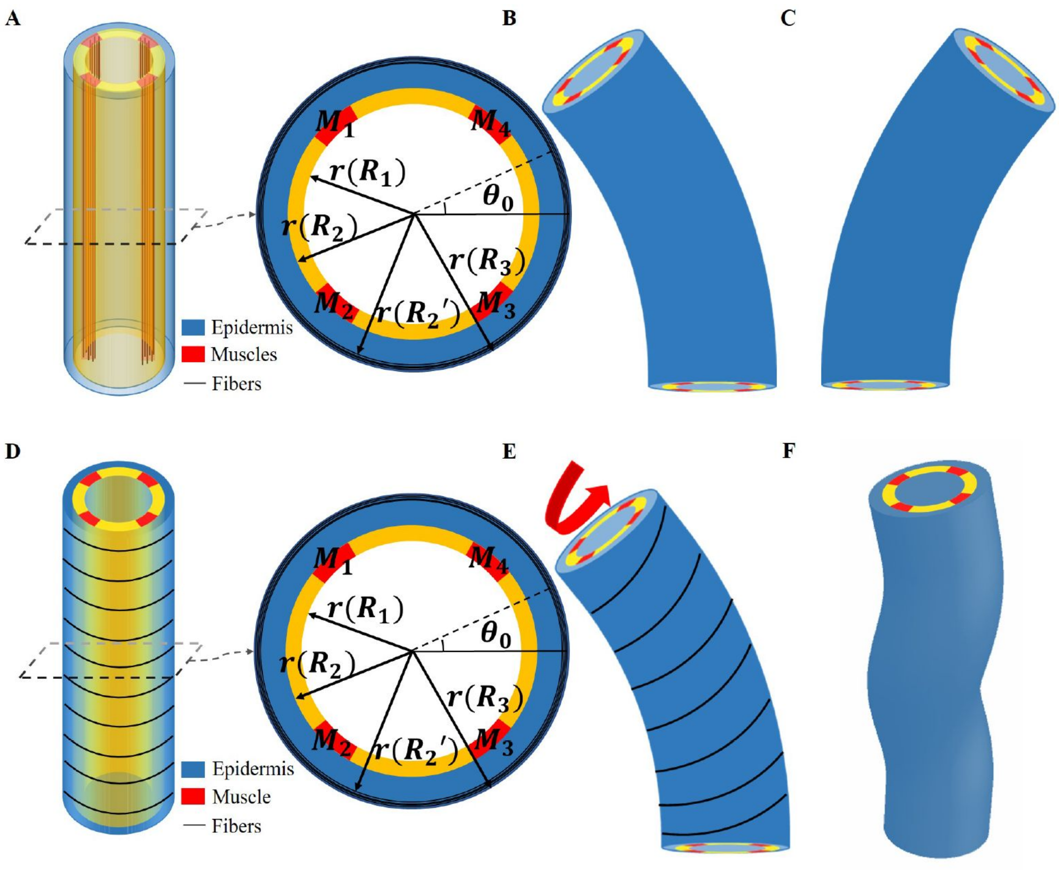


Figure 5.

(A) Schematic diagram of *C. elegans* muscle fibers and its cross section, and it does not show the actin fibers. Four muscle bands exist in the yellow layer. But, the yellow region is not an actual tissue layer and it is simply to define the position of muscles. (B) Deformation diagram, when left side muscles $M_1(\theta_1 \leq \theta_0 \leq \theta_2)$ and $M_2(\theta_3 \leq \theta_0 \leq \theta_4)$ (C) Deformation diagram, when right side muscles $M_3(\theta_5 \leq \theta_0 \leq \theta_6)$ and M_4 $(\theta_7 \leq \theta_0 \leq \theta_8)$. (D) Schematic diagram of *C. elegans* actin fibers and cross-section. (E) Once the muscle is activated, the actin fiber orientation changes from the 'loop' to the 'slope', which results in torque. (F) Schematic diagram of torsional and bending deformation.

this region is activated with a constant strain value, g_a, the angle of the actin fibers will change following the amplitude of the bending by the muscle contraction. In this situation, the angle of the actin fibers may change from $\alpha_a \in \left[0, \frac{\pi}{2}\right]$ but β_a is not modified and β_a = 0.

Throughout the entire process, the muscle and acto-myosin activities are assumed to work almost simultaneously. Our modeling allows us to evaluate the bending and torsion generated independently by muscles and actin bundles, culminating in a complete deformation under coupling. Furthermore, the angle of the acto-myosin fibers varies during muscle contractions. We maintain a constant activation of the acto-myosin network and gradually increase muscle activation.

As a result, the bending of the model will be increased, causing a change in the angle of the actin fibers, ultimately yielding a deformation map that is presented in **Fig.6(A)-(C)**. Upon increasing myosin activation, we observed a consistent torsional deformation (**Fig.6(E)**) that agrees with the patterns seen in the video (**Fig.6(D)**). However, significant torsional deformations are not always present. In fact, other sources can induce torsion as the default of symmetry of the muscle axis, which we will discuss later. In **Fig.6(F)-(G)**, we demonstrate that the curvature provided by the model increases as muscle activation increases and that the torsion is not simply related to the activation amplitude since it also depends on the value of the angle α_a, reaching a maximum at approximately $\pi/4$. Detailed calculations are shown in Section Methods and Materials and **Appendix 4**.

Energy transformation and Elongation

During the late elongation process, the four internal muscle bands cyclically contract in pairs (*Yang (2017)*; *Williams and Waterston (1994)*). Each contraction of a pair increases the energy of the system under investigation, which is then rapidly released to the body. This energy exchange causes the torsion-bending energy to convert into elongation energy, leading to a length increase during the relaxation phase, as shown in **Fig.1** of the **Appendix 5**. With all deformations obtained, **Eq.(2)** can be used to calculate the accumulated energy W_c produced by both the muscles and the acto-myosin activities during a contraction. Subsequently, when the muscles on one side relax, the worm body reverts to its original shape but with a tiny elongation corresponding to the transferred elastic energy. This new state involves the actin network adopting a 'loop' configuration with a strain of εg_{a1} once relaxation is complete. If all accumulated energy from the bending-torsion deformation goes towards elongating the worm body, the accumulated energy W_c and energy W_r following muscle relaxation are equivalent. The activation of actin fibers g_{a1} after muscle relaxation can be calculated and determined by our model.

Once the first stage is well characterized, we can quantify the total energy resulting from both muscles and acto-myosin after each contraction. Then, we used our model to predict elongation in the wild-type *C. elegans*, unc-112(RNAi) mutant and spc-1(RNAi) pak-1(tm403) mutant and further compared the results with experimental observations. The length of the wild type elongates from approximately 90 μm to 210 μm during the muscle-activated phase (*Lardennois et al. (2019)*), the phase lasts about 140 minutes, and the average time interval between two contractions is about 40 seconds (*Yang (2017)*), so the number of contractions can be estimated to be around 210 times. Unfortunately, due to the difficulty to realize quantitative experiments on an embryo always in agitation in the egg shell between bending, torsion, and also rotations around its central axis, one can hypothesize several scenarios: at each step i between state $A_{i,0}$ to $A_{i,1}$ and then $A_{i,2}$, the whole mechanical muscle-myosin energy is transferred to the elongated step $A_{i,2}$, leading to a small $\delta\zeta_i$.

Considering the experimental results shown in **Fig.7(B)**, we determine the optimal values for the activation parameters: $g_m = -0.15$ and $g_a = -0.01$ assuming that all the accumulated energy during muscle activation is transferred to elongation($W_r = W_c$). The elementary elongation $\delta\zeta_i$ will be gradually increased with time, which is shown as the black line in **Fig.7(A)**. At the beginning,

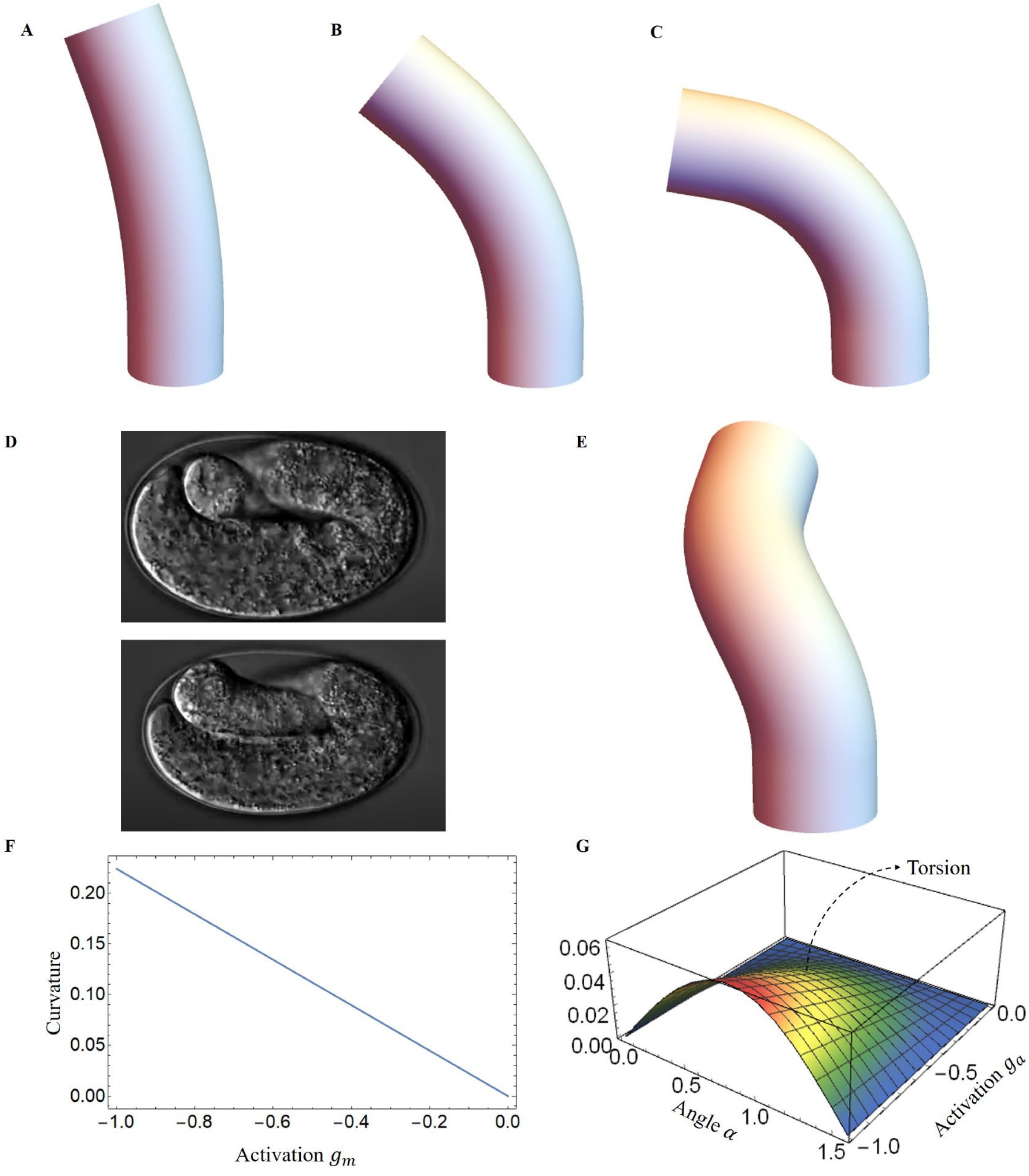


Figure 6.

Deformed configurations for different activation for muscles, (A) $g_m = -0.02$, $\alpha_a = \pi/3$, $g_a = -0.01$. (B) $g_m = -0.05$, $\alpha_a = \pi/4$, $g_a = -0.01$. (C) $g_m = -0.08$, $\alpha_a = \pi/6$, $g_a = -0.01$. (D) The graphs were captured from the Hymanlab, and the website: *https://www.youtube.com/watch?v=M2ApXHhYbaw*. The movie was acquired at a temperature of 20^0C using DIC optics. (E) $g_m = -0.1$, $\alpha_a = \pi/4$, $g_a = -0.7$. (F) Curvature is plotted as a function of muscle activation. (G) Torsion is plotted as a function of the actin activation and angle of actin fibers.

$\delta\zeta_i$ is about 0.5μm, but at the end of this process, $\delta\zeta_i$ is about 1.5μm, indicating that the worm will elongate up to 290μm. The result is significantly higher than our actual size 210μm. When the elongation proceeds, we assume a transfer of energy between bending-torsion-contraction and elongation but it may be not fully effective, which means a significant part of the energy is lost. From the experimental data, we evaluate that the energy loss gradually increases, from full conversion at the beginning to only 40% of the accumulated energy used for elongation at the end of the process ($W_r = 0.4W_c$). It induces for $\delta\zeta_i$ a first increase and then a decrease, which is shown as the blue line in **Fig.7(A)** and is responsible for the slowdown around 200 mins. This option, which may be not the only possible one, leads to the estimated elongation having a good agreement with experimental data (see the blue-dashed curve in **Fig.7(B)**). Indeed it is possible that the *C. elegans* elongation requires other transformations which will cost energy. As the embryo gradually elongates, energy dissipation and the biomechanical energy required to reorganize the actin bundles may be two factors that contribute to the increased energy loss that underlies the hypothesis.

Moreover, it was reported in (***Norman and Moerman (2002***)) that the knockdown of unc-112(RNAi), known for impairing muscle contractions, results in the arrest of elongation of embryos at the twofold stage, indicating that muscles have no activation, $g_m = 0$ in our model, and no accumulated energy can be converted into elongation. Another mutation concerning the embryos consisting of mutant cells with pak-1(tm403), known to regulate the activity of myosin motors leads to a retraction of the embryo, so the pre-stretch caused by myosin will not be maintained and will decrease. These aforementioned findings are fully consistent with a variety of experimental observations and are shown in **Fig.7(B)**.

Embryo rotations and dissipation

The main manifestation of the muscle activity, independently of the elongation, is probably the constant rotations of the embryo despite its confinement in the egg. This can be explained by a small angular deviation of the muscle sarcomeres from the central axis due to their attachment to the inner boundary of the cell epidermis, the so-called "dense bodies" (***Moerman and Williams (2006***)). Since they cross the horizontal plane at approximately $\pm 45^\circ$ and the deviation β_m from the anterior-posterior axis is estimated to be about 6° each active muscle on the left (or on the right) contributes to the torque via a geometrical factor about $a_g = \sin(6\pi/180)\cos(\pi/4)$. Then a simple estimation of the muscle activity in terms of torque reads $\Lambda_m \sim \mu_m \pi R^3 s_m p_m a_g (\epsilon g_m)$ where s_m is the surface of the muscle pair on the left (or right) compared to the section of the cylinder: $s_m = 0.025$ and p_m is the distance of the muscles from the central axis of the embryo: $p_m = 0.75$ while $g_m = 0.15$ according to the analysis of the elongation. So the muscles on one side contribute to a torque Λ_m along the axis of symmetry given by $\Lambda_m = 4.657\mu_m \pi R^3 \cdot 10^{-5}$.

Let us consider now the dissipative torque, assuming that the dynamics of rotation is stopped by friction after one bending event. Two cases can be considered: either the dissipation comes from viscous flow or from the rubbing of the embryo when it folds. The fluid dissipation results from the rotations in the interstitial fluid inside and along the egg shell since the anterior-posterior axis remains parallel to the egg-shell (***Bhatnagar et al. (2023***)) The interstitial fluid, of viscosity η contains a significant amount of sugar and other molecules which are required for embryo survival and then is more viscous than water (***Labouesse (2023***)). However, values for sucrose or sorbitol at the concentration of 1 mole/liter indicates a viscosity of order a few times the viscosity of water, which is 1 mPas. For example at 0.9 mole/liter and temperature of 20°, an aqueous solution with sorbitol has a viscosity of 1.6 mPas, which can be extrapolated to $\eta = 1.9$ mPas at 1.2 mole/liter (***Jiang et al. (2013***)). The estimation given by an embryo located in the middle of the egg-shell, gives a weaker viscous torque once evaluated by $\Lambda_v = 4\pi\eta\Omega_e L R^2 \left(R_{egg}^2/(R_{egg}^2 - R^2)\right)$ according to a classical result reported by Landau *et al.* (***Landau and Lifshitz (1971***, 2013)). It has to be mentioned that this estimation assumes that the two cylinders: the egg shell and the embryo have the same axis of symmetry and concerns the beginning of the muscle activity where the

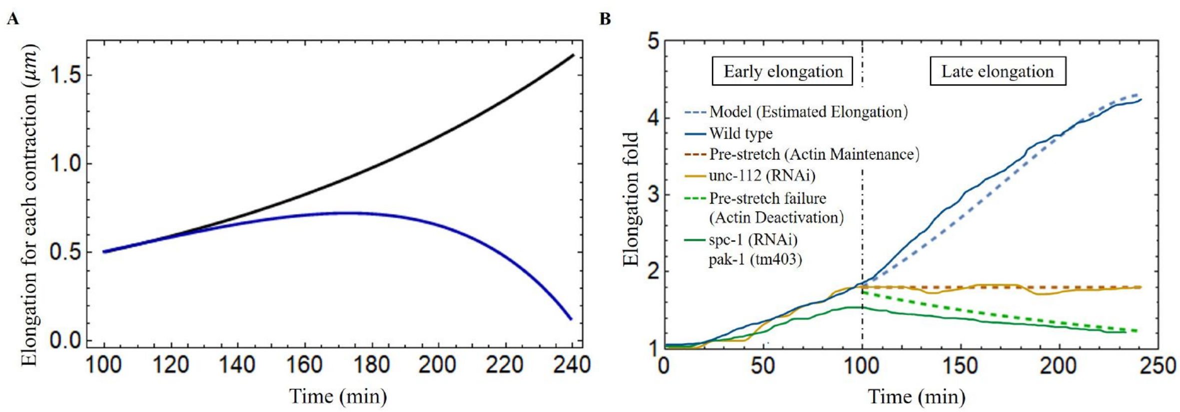


Figure 7.

(A) The elongation for each contraction varies with time. Black line: all energy converted to the elongation, blue line: partial energy converted to the elongation. The activation: g_m = −0.15, g_a = −0.01. (B) The model predicted results agree well with the experimental data of wild-type and different mutant *C. elegans* embryos (***Lardennois et al. (2019***)). The activation of wild type model (blue dashed line): g_m = −0.15, g_a = −0.01. The activation of unc-112(RNAi) (brown dashed line): g_m = 0, g_a = 0. In the pre-stretch failure case (green dashed line), λ will decrease from 1.8.

radius is about 8.2 μm, the length is 90 μm,the radius of the shell about $R_{egg} = 15\mu m$ and the length $L_{egg} = 54\mu m$, see **Fig.6(D)**. The angular velocity Ω_e is more difficult to evaluate but it is about 90° per two seconds deduced from videos. When the embryo approaches the egg shell, the friction increases, and two eccentric cylinders of different radii have to be considered with the two axes of symmetry separated by a distance d. The hydrodynamic study in this case is far from being trivial, and seems to have been initiated first by Zhukoski (***Zhukoski (1887***)) who suggested the use of bipolar coordinates for the mathematical treatment. Many following contributions established with different simplified assumptions have been published after and the study was fully revisited by Ballal and Rivlin (***Ballal*** and Rivlin (***1976***)). Here we focus on the limit of a small gap δ between the rotating body-shape and the egg and by considering an asymptotic analysis at small $\underline{\delta}$ of the general result derived in (***Ballal and Rivlin (1976***)). Thus, the viscous torque reads $\tilde{\Lambda}_v = 2\sqrt{2}\pi\eta\Omega_e L_c R_{egg}^2 \sqrt{R_{egg}/\delta}\sqrt{(R_{egg}-d)/d}$ is the zone of contact with the egg and $d = R_{egg} - \delta - R$. This approach is an approximation since the embryo has more of a torus shape than a cylinder (***Yang Xinyi et al. (2023***)) but the evaluation of the dissipation is satisfactory for $\delta = 0.5\mu m$, $\Omega_e \sim \pi/4\ s^{-1}$ and $\mu = 10^5 P\ a$. Coming back to the first model of dissipation with the same data, the ratio between the dissipative viscous torque and the active one gives: $\Lambda_v/\Lambda_m = 0.02$, which is obviously unsatisfactory. Finally, the dissipative energy ? during one bending event leading to an angle of $\pi/2$ is $\xi_{diss} = 1/2\Lambda_m \times(\pi/2)^2$ which represents 4% of the muscle elastic energy during the bending so at the beginning of the muscle activity (Appendix 5, Eq.(45)), the dissipation exits but is negligible. At the very end of the process, this ratio becomes 60% but as yet mentioned our estimation for the dissipation becomes very approximate, increases a lot due to the embryo confinement, and does not involve the numerous biochemistry steps necessary to reorganize the active network: acto-myosin and muscles.

Discussion

Since the discovery of the muscle activity before the egg hatch of the *C. elegans* embryo, it has become critical to explain the role of mechanical forces generated by muscle contraction on the behavioral and functional aspects of the epidermis. We provide a mechanical model in which the *C. elegans* is simplified as a cylinder, and the muscle bands and actin that drive its elongation are modeled as active structures in a realistic position. We determine the fiber orientation using experimental observations and then calculate the deformation by tensorial analysis involving the strains generated by the active components. Although a special focus is made on late elongation, its quantitative treatment cannot avoid the influence of the first stage of elongation due to the acto-myosin network, which is responsible for a pre-strain of the embryo. In a finite elasticity formalism, the deformations induced by muscles in a second step are coupled to the level of strains of the initial elongation period. For that, we need to revisit the theory of the acto-myosin contraction and previous results (***Ciarletta et al. (2009***); ***Vuong-Brender et al. (2017a***); ***Ben Amar et al. (2018***)) to unify the full treatment. In particular, a model for the recruitment of active myosin motors under forcing is presented which recovers the experimental results of the first elongation stage.

The elongation process of *C. elegans* during the late period is much more complex than the early elongation stage which is caused only by actin contraction. During the late elongation, the worm is distorted by the combined action of muscle and acto-myosin, resulting in an energy-accumulating process. Bending deformation is a phenomenon resulting from unilateral muscular contraction, and during the late elongation, significant torsional deformation is observed, indicating that the bending process induces a reorientation of the actin fibers. It is worth noting that the embryo is always rotating in tandem with the muscle activity making difficult any experimental measurements. However, our model can predict that if the muscles are not perfectly vertical, torque exits that causes rotation and eventually torsion. The accumulated energy is then partly turned into energy for the ongoing action of actin, allowing the embryo to elongate when the muscle relaxes. Both sides of the *C. elegans* muscles contract in a sequential cycle, repeating the energy conversion process, and eventually completing the elongation process. However, the energy exchange between bending and elongation is limited, among other factors, by the viscous dissipation induced by rotation, which is also evaluated in this study. Not investigated in detail here is the necessary re-organization of the active networks (acto-myosin and muscles) due to this tremendous shape transformation of the embryo. In parallel to elongation, the cuticle is built around the body (***Page*** and Johnstone (***2007***)). This very thin and stiff membrane ensures protection and locomotion post-hatching. Clearly, these processes will perturb muscle activity. These two aspects which intervene in the final stage of the worm confinement play a very important role at the frontier across scales between genetics, biochemistry and mechanics.

Finally, the framework presented here not only provides a theoretical explanation for embryonic elongation in *C. elegans*, it can also be used to model other biological behaviors, such as plant tropism (***Moulton et al. (2020b***)) and elephant trunk elongated (***Schulz et al. (2022b***,a)) and bending. Our ideas could potentially be used in the emerging field of soft robotics, like octopus legs-inspired robots (***Kang et al. (2012***); ***Nakajima et al. (2013***); ***Calisti et al. (2015***)), which is soft and its deformation induced by muscles activation. We can reliably predict deformation by knowing the position of activation and the magnitude of forces in the model. Furthermore, residual stresses can be incorporated into our model to fulfill design objectives.

Methods and Materials

The model has been presented in a series of articles by A. Goriely and collaborators (***Goriely and Ta-*** bor (***2013***); ***Moulton et al. (2013***, 2020b); ***Kaczmarski et al. (2022***); ***Goriely et al. (2022***)). As one imagines, it is far from triviality and most of the literature on this subject concerns either full cylinders or cylindrical shells in torsion around the axis of symmetry, the case of bending is more overlooked. However, the geometry of the muscles in *C. elegans* leads automatically to a bending process that cannot be discarded. We take advantage of these previous works and apply the methodology to our model.

The deformation gradient

The axial extension ζ is obtained by $\mathbf{r}'(Z) = \zeta\, \mathbf{d}_3$, where $'$ denotes the first derivative with respect to the material coordinate Z. From the director basis, the Darboux curvature vector reads: $\mathbf{u} = u_1\mathbf{d}_1 + u_2\mathbf{d}_2 + u_3\mathbf{d}_3$, this vector gives the evolution of the director basis along the filamentary line as: $\mathbf{d}_i'(Z) = \zeta\, \mathbf{u} \times \mathbf{d}_i$, see **Fig.3(A)**.

The finite strain $\mathbf{G}_0$ can map the initial stress-free state $\mathscr{B}_0$ to a state $\mathscr{B}_1$, which reflects early elongation process. After in the residually stressed $\mathscr{B}_1$, we impose an incremental strain field $\mathbf{G}_1$ maps the body to $\mathscr{B}_2$, which represents late elongation process. So, the deformation gradient can be expressed as: $\mathbf{F} = \mathbf{F}_e\mathbf{G}_1\mathbf{G}_0$ (***Goriely and Ben Amar (2007***)). The deformation gradient $\mathbf{F} = F_{ij}\, \mathbf{d}_i \otimes \mathbf{e}_j$, $i \in \{1, 2, 3\}$ and j labels the reference coordinates $\{R, \Theta, Z\}$ finally reads:

$$\mathbf{F} = \begin{bmatrix} a_{1R} & \frac{1}{R}a_{1\Theta} & \lambda\varepsilon\left(1+\varepsilon\xi\right)\left(u_2 a_3 - u_3 a_1 \sin a_2\right) \\ a_1 a_{2R} & \frac{a_1}{R}a_{2\Theta} & \lambda\varepsilon\left(1+\varepsilon\xi\right)\left(u_3 a_1 \cos a_2 - u_1 a_3\right) \\ a_{3R} & \frac{1}{R}a_{3\Theta} & \lambda\left(1+\varepsilon\xi\right)\left(1+\varepsilon\left(u_1 a_1 \sin a_2 - u_2 a_1 \cos a_2\right)\right) \end{bmatrix}, \tag{4}$$

where λ is the axial extension due to the pre-strained.

We define $\mathbf{G}_0 = Diag\,(1, 1+\varepsilon c, 1)$, since $g_0 = 0.88$ can be determined through the early elongation, we write it in the form of $1 + \varepsilon c$ to simplify the calculation, $c \approx -0.6$ ($\varepsilon \approx 0.2$).

Let us consider fibers oriented along the unit vector $\mathbf{m}$ = (sin α sin β, sin α cos β, cos α) for the incremental strain $\mathbf{G_1}$, where α and β characterise the angles with $\mathbf{e_z}$ and $\mathbf{e}$ and can be found in **Fig.3(A)**. The active filamentary tensor in cylindrical coordinates is then $\mathbf{G} = \mathbf{G}_0\,(\mathbf{I} + \varepsilon g\,\mathbf{m} \otimes \mathbf{m})$, see (***Holzapfel (2002***)) which reads:

$$\mathbf{G} = \mathbf{G_0} + \varepsilon g \mathbf{G_0} \begin{bmatrix} \sin^2\alpha \sin^2\beta & \sin^2\alpha \sin\beta\cos\beta & \sin\alpha\cos\alpha\sin\beta \\ \sin^2\alpha\sin\beta\cos\beta & \sin^2\alpha\cos^2\beta & \sin\alpha\cos\alpha\cos\beta \\ \sin\alpha\cos\alpha\sin\beta & \sin\alpha\cos\alpha\cos\beta & \cos^2\alpha \end{bmatrix}. \quad (5)$$

The energy function

The auxiliary strain-energy density can be obtained by expanding the inner variables $a_i\ p_i$ and the potential V (see Eq.(2)) as:

$$\mathbf{a} = \mathbf{a}^{(0)} + \varepsilon \mathbf{a}^{(1)} + \varepsilon^2 \mathbf{a}^{(2)} + \mathcal{O}\left(\varepsilon^3\right), \quad (6)$$

$$p = p^{(0)} + \varepsilon p^{(1)} + \varepsilon^2 p^{(2)} + O\left(\varepsilon^3\right), \quad (7)$$

$$V(\mathbf{F}_e, \mathbf{G}) = V_0 + \varepsilon V_1 + \varepsilon^2 V_2 + O\left(\varepsilon^3\right). \quad (8)$$

For each cross-section, the associated Euler-Lagrange equations take the following form:

$$\frac{\partial}{\partial R}\frac{\partial V_i R}{\partial a_{jR}^{(k)}} + \frac{1}{R}\frac{\partial V_i R}{\partial a_{jR}^{(k)}} + \frac{\partial}{\partial \Theta}\frac{\partial V_i R}{\partial a_{j\Theta}^{(k)}} - \frac{\partial V_i R}{\partial a_j^{(k)}} = 0, \quad (9)$$

where j = 1, 2, 3, k = 0, 1, in association with boundary conditions at each order k = 0, 1 on the outer radius R_i:

$$\left.\frac{\partial V_i R}{\partial a_{jR}^{(k)}}\right|_{R=R_i} = 0, j = 1, 2, 3. \quad (10)$$

We solve these equations order by order and consider incompressibility. At the lowest order, the incompressibility imposes the deformation: $\mathbf{a}^{(0)} = (r\,(R), \Theta, 0)$ and the Euler-Lagrange equation gives the Lagrange parameter $p^{(0)} = P_0(R)$ which finally read:

$$r' = \frac{R}{\lambda r},\ \ P_0{}' = 0, \quad (11)$$

and the boundary condition $\sigma_{rr}(R = 1) = 0$ gives the value of $P_0 = 1/\lambda$. At $\mathcal{O}\,(\varepsilon)$, the Euler-Lagrange equations are again automatically satisfied, and at $\mathcal{O}(\varepsilon^2)$, the crucial question is to get the correct expression for $a_i^{(1)}$ and $p(1)$. Based on the previous subsection, the following form is instuited.

$$\begin{aligned} a_1^{(1)} &= -\frac{1}{2} r\,(R)\,\xi + q_1 r(R)^2 \left(u_1 \sin\Theta - u_2\cos\Theta\right) + h_1\,(R) \\ a_2^{(1)} &= q_2 r\,(R)\left(u_1\cos\Theta + u_2\sin\Theta\right) + h_2\,(R) \\ a_3^{(1)} &= \lambda\xi \\ p^{(1)} &= h_3\,(R) \end{aligned} \quad (12)$$

As before, the Euler-Lagrange equations and incompressibility condition give the two constants $q_1 = \frac{1}{8}\left(-2 + \lambda^3\right)$, $q_2 = \frac{1}{8}\left(2 + 3\lambda^3\right)$ and h_1, h_2, h_3 are a function of R. With the solutions for active strain $\mathbf{a}^{(0)}$ and $\mathbf{a}^{(1)}$, full expressions of them are given in the Appendix 4. The second order energy takes the form:

$$\mathcal{E} = \varepsilon^4 \int_0^L \mathrm{d}Z \int_S V_2\left(\mathbf{a}^{(0)}, \mathbf{a}^{(1)}; \mathrm{u}_1, \mathrm{u}_2, \mathrm{u}_3, \xi\right) R\mathrm{d}R\ \mathrm{d}\Theta + \mathcal{O}\left(\varepsilon^5\right), \quad (13)$$

and we can transform it into this form:

$$V_2 R = A_1 \xi + A_2 \xi^2 + B_1 u_1 + B_2 {u_1}^2 + C_1 u_2 + C_2 {u_2}^2 + D_1 u_3 + D_2 {u_3}^2, \tag{14}$$

where A_i, B_i, C_i and D_i are functions of R and Θ.

The method for determining deformation

Comparing with the energy of an extensible elastic rod (***Moulton et al. (2020a)***; ***Kirchhoff and*** Hensel (***1883***); ***Mielke (1988***); ***Mora and Müller (2002***, 2004)), we recognize the classic extensional stiffness K_0, bending stiffness K_1 and K_2, torsional stiffness K_3 coefficients:

$$K_0 = \int_{S_K} A_2 d\,R d\Theta,\ K_1 = \int_{S_K} B_2 d\,R d\Theta,\ K_2 = \int_{S_K} C_2 d\,R d\Theta,\ K_3 = \int_{S_K} D_2 d\,R d\Theta, \tag{15}$$

where A_2, B_2, C_2 and D_2 are related to the shear modulus μ, so for a uniform material with no variations of shear modulus, S_K represents the cross-section of the cylinder. But if not, we need to divide different regions to perform the integration.

We now focus on the intrinsic extension and curvatures of the cylindrical object induced by the active strains, this requires the competition of the active forces with the stiffness of the cylindrical bar following the relationships,

$$\hat{\zeta} = 1 - H_0/K_0,\ \hat{u}_1 = -H_1/K_1, \hat{u}_2 = H_2/K_2,\ \hat{u}_3 = -H_3/K_3, \tag{16}$$

where H_i calculated by:

$$H_0 = \frac{1}{2}\int_{S_H} A_1 d\,R d\Theta,\ H_1 = \frac{1}{2}\int_{S_H} B_1 d\,R d\Theta,\ H_2 = \frac{1}{2}\int_{S_H} C_1 d\,R d\Theta,\ H_3 = \frac{1}{2}\int_{S_H} D_1 d\,R d\Theta. \tag{17}$$

where A_1, B_1, C_1 and D_1 are related to the shear modulus μ,the fiber angles α and β, and the activation g. So, the integration region S_H is divided into the different parts of the embryo which all contribute to the deformation.

To calculate the intrinsic Frenet curvature and torsion, we use the following equation:

$$\hat{\kappa} = \sqrt{\hat{u}_1^2 + \hat{u}_2^2} \quad \text{and} \quad \hat{\tau} = \frac{\hat{u}_3}{\hat{\zeta}}. \tag{18}$$

Finally, for the case of activated left side muscles, we can calculate intrinsic extension and curvatures by Eq.16), $\hat{u}_{1m} = 0$ and $\hat{u}_{3m} = 0$, so the intrinsic Frenet curvature and torsion Eq.(18), $\hat{\kappa}_m = \hat{u}_{2m}$ and $\hat{\tau}_m = 0$. For the activated actin case, we can calculate intrinsic extension and curvatures by Eq.(16), $\hat{u}_{1a} = 0$ and $\hat{u}_{2a} = 0$, and the intrinsic Frenet curvature and torsion Eq.(18), $\hat{\kappa}_a = 0$ and $\hat{\tau}_a = \frac{\hat{u}_{3a}}{\hat{\zeta}_a}$. These quantities have been used to calculate the deformation images, as shown previously in **Fig.6**.

After obtaining all solutions, we can use these quantities to calculate the accumulated energy W_c in the system after one contraction by **Eq.(2)**. By defining the activation (g_m and g_a) and the conversion efficiency of energy, we can obtain the activation (g_{a1}) of the actin bundles after once contraction and further calculate the elongation.

All calculation results and parameters are presented in the Appendix.

Acknowledgements

It is a pleasure to acknowledge fruitful discussions with Michel Labouesse and Kelly Molnar about experimental observations concerning the embryonic *C.elegans* elongation. We also thank Alain Goriely for his help with the technical aspects of nonlinear elasticity. Both authors acknowledge the support of the contract EpiMorph (ANR-2018-CE13-0008). Anna Dai acknowledges the support of the CSC (China Scholarship Council), file No. 201906250173.

Appendix 1

Size and material parameters

Geometry parameters	Normalized radius[1]	R_1=0.7	R_2=0.8	R_2'=0.96	R_3=1
	The location of muscles[2]	$\theta_1 = \frac{2}{3}\pi$ $\theta_2 = \frac{5}{6}\pi$	$\theta_3 = \frac{7}{6}\pi$ $\theta_4 = \frac{4}{3}\pi$	$\theta_5 = \frac{5}{3}\pi$ $\theta_6 = \frac{11}{6}\pi$	$\theta_7 = \frac{1}{6}\pi$ $\theta_8 = \frac{1}{3}\pi$
Material parameters (Shear modulus)[3]	Actin part	$\mu_a = 5$			
	Epidermis part	$\mu_e = 1$			
	Muscles	$\mu_m = 100$			
	Soft inner part	$\mu_i = 1/200$			

[1] The radius are extracted or deduced by the Ref.(***Ben Amar et al.*** (***2018***); ***Lardennois et al.*** (***2019***)).
[2] The location and size of muscles are deduced by the Ref.(***Moerman and Williams*** (***2006***); ***Lardennois et al.*** (***2019***)).
[3] The units of the shear modulus are the KPa and is scaled by the epidermis one. It gives for the muscle, a value consistent with the muscle shear modulus proposed in (***Denham et al.*** (***2018***)).

Appendix 1–table 1.

Parameters adopted in this work

Appendix 2

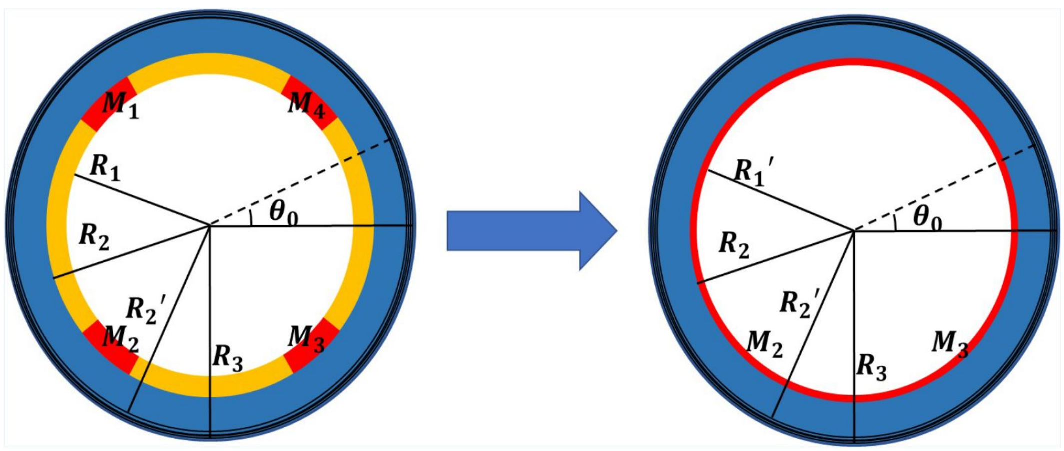
Analytical model of the early elongation

The finite deformation gradient: $\mathbf{F}_0 = Diag(r'(R), r(R)/R, \lambda)$, and $\mathbf{G}_0 = Diag(1, g_0, 1)$, $0 < g_0 < 1$ in the actin layer, and $g_0 = 1$ in the part without actin budles. To determine the finite strain $\mathbf{G}_0$, we converted the four muscle parts into thin layers attached to the epidermis in equal proportions to ensure the continuity of the model, and divide our model into four parts, see **Fig.1**. The actin layer($R_3 < R < R_2'$), epidermis layer($R_2 < R < R_2'$) and muscle layer($R_1' < R < R_2$) are considered as incompressible materials, and the inner part($0 < R < R_1'$) is compressible. The Neo-Hookean energy function is used for the incompressible parts:

$$W = \mu \left[\frac{1}{2} \left(Tr \left(\mathbf{F_e F_e}^T \right) - 3 \right) - p \left(\det \mathbf{F_e} - 1 \right) \right]. \quad (19)$$

According to the Euler-Lagrange equations, we can obtain the radius of actin layer(r_a), epidermis layer(r_e) and muscle layer(r_m):

$$r_a = \sqrt{\frac{g_0 R^2 + A}{\lambda}}, \quad r_e = \sqrt{\frac{R^2 + E}{\lambda}}, \quad r_m = \sqrt{\frac{R^2 + M}{\lambda}}. \quad (20)$$



Appendix 2–figure 1.

Cross-sectional simplified model with four scattered muscle sections simplified to thin layers ($R_1^{'}$ = 0.768).

where A, E and M are constant, and the Lagrange multiplier p in the actin, epidermis and muscle layer:

$$p_a = \frac{-\log(R) + \frac{1}{2}{g_0}^2\left(-\frac{A}{A+g_0R^2} + \log\left(A+g_0R^2\right)\right)}{g_0\lambda} + C_a, \tag{21}$$

$$p_e = -\frac{\frac{E}{E+R^2} + 2\log(R) - \log\left(E+R^2\right)}{2\lambda} + C_e, \tag{22}$$

$$p_m = -\frac{\frac{M}{M+R^2} + 2\log(R) - \log\left(M+R^2\right)}{2\lambda} + C_m. \tag{23}$$

For the compressible part, we take the energy function form (***Holzapfel (2002***)):

$$W = \frac{\mu}{2}\left(Tr\left(\mathbf{F_e}{\mathbf{F_e}}^T\right) - 3 + \kappa\left(\det \mathbf{F_e} - 1\right)^2\right) \tag{24}$$

where κ is a material constant. The radius of the inner part:

$$r_i = \frac{aR}{\sqrt{\lambda}} \tag{25}$$

where a is a constant.

By considering the boundary condition $\sigma_{rr} = 0$ on the outer border $R = 1$:

$$\sigma_{arr}(R=1) = \mu_a\left[\left(\frac{\partial r_a(1)}{\partial R}\right)^2 - p_a(1)\right] = 0 \tag{26}$$

the continuity of the radius:

$$r_a\left(\mathrm{R_2}'\right) = r_e\left(\mathrm{R_2}'\right), \quad r_e\left(\mathrm{R_2}\right) = r_m\left(\mathrm{R_2}\right), \quad r_m\left(\mathrm{R_1}'\right) = r_i\left(\mathrm{R_1}'\right), \tag{27}$$

and the continuity of the radial stresses σ_{rr} in $R_2{}'$ and R_2 :

$$\mu_a\left[\left(\frac{\partial r_a\left(\mathrm{R_2}'\right)}{\partial R}\right)^2 - p_a\left(\mathrm{R_2}'\right)\right] = \mu_e\left[\left(\frac{\partial r_e\left(\mathrm{R_2}'\right)}{\partial R}\right)^2 - p_e\left(\mathrm{R_2}'\right)\right], \tag{28}$$

$$\mu_e\left[\left(\frac{\partial r_e\left(\mathrm{R_2}\right)}{\partial R}\right)^2 - p_e\left(\mathrm{R_2}\right)\right] = \mu_m\left[\left(\frac{\partial r_m\left(\mathrm{R_2}\right)}{\partial R}\right)^2 - p_m\left(\mathrm{R_2}\right)\right]. \tag{29}$$

According to the above equations, we can obtain expressions for constants E, M, C_a, C_e and C_m.

We substitute all size, material parameters and $\kappa = 100$, $\lambda = 1.8$ into the last condition σ_{mrr} ($\mathrm{R_1}'$)= σ_{irr} ($\mathrm{R_1}'$):

$$\mu_m\left[\left(\frac{\partial r_m\left(\mathrm{R_1}'\right)}{\partial R}\right)^2 - p_m\left(\mathrm{R_1}'\right)\right] = \mu_i\sigma_{irr}\left(\mathrm{R_1}'\right), \tag{30}$$

where σ_{irr}:

$$\sigma_{irr} = \frac{2}{\det \mathbf{F_e}}\left((\det \mathbf{F_e})^2 \frac{\partial W}{\partial (\det \mathbf{F_e})^2} + \frac{\partial W}{\partial (tr(\mathbf{F_e}\mathbf{F_e}^T))} r_i'(R)^2\right). \quad (31)$$

and obtain the relationship with constant A and g_0.

Finally, by prescribing a zero traction condition on the top of the cylinder, noticeably, the muscle part was considered inextensible, so no stress on the top:

$$\int_{R_2'}^{1} \sigma_{azz} r_a r_a' dR + \int_{R_2}^{R_2'} \sigma_{ezz} r_e r_e' dR + \int_{0}^{R_1'} \sigma_{izz} r_i r_i' dR, \quad (32)$$

where:

$$\sigma_{azz} = \lambda^2 - p_a, \quad \sigma_{ezz} = \lambda^2 - p_e, \quad (33)$$

$$\sigma_{izz} = \frac{2}{\det \mathbf{F_e}}\left((\det \mathbf{F_e})^2 \frac{\partial W}{\partial (\det \mathbf{F_e})^2} + \frac{\partial W}{\partial (tr(\mathbf{F_e}\mathbf{F_e}^T))} \lambda^2\right), \quad (34)$$

all solutions g_0 = 0.88, A = 0.08 and $r_a(1)$ = 0.73 can be determined. The result is also in good agreement with the experimental data (***Vuong-Brender et al. (2017a***); ***Ben Amar et al. (2018***)).

Appendix 3

Time-scale for myosin detachment

The time required for non-muscle myosin detachment is estimated to be $\tau_0 = 10s$ for free acto-myosin filaments. If the actin filament is submitted to external loading perpendicular to its axis, the detachment can be helped or on the contrary inhibited, see (***Howard and*** Clark (***2002***)), page 169-170. In the present case, the stresses acting in the radial direction of an actin bundle are compressive and thus will delay the detachment. This energy has to be compared to the energy of detachment of all myosin motors from the bundle. The corresponding elastic energy associated with the radial deformation for an actin cable of length l_a of radius $r_b = 0.05\mu m$ estimated from (***Lardennois et al. (2019***)) and shear modulus $\mu_a = 5\ KP\ a$ is given by $1/2\mu_a(1-g_0(t))\pi l_a r_b^2 = 210^{-11} l_a\ J$. This result must be compared to the individual energy of detachment times the number of myosin motors on a cable. This number is uneasy to fix but estimation is given by the length of the cable l_a divided by the distance between 2 anchoring sites of myosin which is about $5nm$ while the attachment energy per motor is about $6k_bT$. These indications are for skeletal muscle myosins (***Howard and*** Clark (***2002***)) and have to be taken with caution. Nevertheless, the order of magnitude of the debonding energy for a collection of myosin heads from actin cable can be estimated to $4.8 l_a 10^{-12} J$. Then the ratio between both quantities is of order $4(1-g_0(t))$ which explains that the time-scale of debonding for a cable in compressive stress in the orthogonal direction of its axis is then:

$$\tau_{deb} = \tau_0 e^{p_3(1-g_0(t))} \quad (35)$$

where p_3 is a positive constant of order one which cannot be predicted exactly. This time scale justifies the exponential correction in Eq.(3) of the manuscript.

Appendix 4

Modelling details of without pre-strain case

In the paper, we discuss the case with pre-strain. Here, more details about without prestrain ($\mathbf{G_0} = \mathbf{I}$) case are displayed. To lowest order, the solution of the Euler–Lagrange equations is obviously given by $\mathbf{a}^{(0)} = (R, \Theta, 0)$, and $p^{(0)} = 1$ so there is no deformation and the zero and first order of energy vanish. At order $\mathcal{O}(\epsilon^2)$ of the elastic energy, the solution for the active strain components are:

$$
\begin{aligned}
a_1^{(1)} &= -\frac{R}{2}\xi - \frac{R^2}{8}\left(u_1 \sin\Theta - u_2 \cos\Theta\right) + f_1(R),\\
a_2^{(1)} &= \frac{5R}{8}\left(u_1 \cos\Theta + u_2 \sin\Theta\right) + f_2(R),\\
a_3^{(1)} &= 0,\\
p^{(1)} &= f_3(R).
\end{aligned} \tag{36}
$$

where $f_{1,2}$ are functions related to the active stress g and fiber angles α and β:

$$
\begin{aligned}
f_1(R) &= \frac{gR}{2},\\
f_2(R) &= g log(R) \sin^2\alpha \sin(2\beta),\\
f_3(R) &= 2g\mu log(R) \sin^2\alpha \sin(2\beta).
\end{aligned} \tag{37}
$$

The coefficients for determining stiffness:

$$
\begin{aligned}
A_2 &= \frac{9\mu R}{8},\\
B_2 &= \frac{81}{128}\mu R^3 \sin^2\Theta,\\
C_2 &= \frac{81}{128}\mu R^3 \cos^2\Theta,\\
D_2 &= \frac{\mu R^3}{2}.
\end{aligned} \tag{38}
$$

The coefficients for determining deformation:

$$
\begin{aligned}
A_1 &= -\frac{1}{8} g\mu R\left[4 + 5\cos(2\alpha) + 2\cos(2\beta)(-1 + 2\log(R))\sin^2\alpha\right],\\
B_1 &= -\frac{1}{16} g\mu R^2\left[10 + 17\cos(2\alpha) + 2\cos(2\beta)(-1 + 14\log(R))\sin^2\alpha\right]\sin\Theta,\\
C_1 &= \frac{1}{16} g\mu R^2\left[10 + 17\cos(2\alpha) + 2\cos(2\beta)(-1 + 14\log(R))\sin^2\alpha\right]\cos\Theta,\\
D_1 &= -g\mu R^2 \cos\beta \sin(2\alpha).
\end{aligned} \tag{39}
$$

Modelling details of with pre-strain case

The first order solutions of the theory with the pre-strain case, $h_{1,2,3}(R)$ are related to activation g and fiber angles α and β:

$$
\begin{aligned}
h_1(R) &= \frac{R(c+g)}{2\sqrt{\lambda}},\\
h_2(R) &= g\log(R)\sin^2\alpha\sin(2\beta),\\
h_3(R) &= \frac{2\mu log(R)\left[c + g\cos(2\beta)\sin^2\alpha\right]}{\lambda}.
\end{aligned} \tag{40}
$$

The coefficients for determining stiffness:

$$
\begin{aligned}
A_2 &= (0.278 + 1.689\mu)\, R, \\
B_2 &= 1.294 R^3 \left[-0.405 + 2.389\mu + (-0.176 + \mu)\cos(2\Theta)\right], \\
C_2 &= -1.294 R^3 \left[0.405 - 2.389\mu + (-0.176 + \mu)\cos(2\Theta)\right], \\
D_2 &= 0.9\mu R^3.
\end{aligned}
\tag{41}
$$

The coefficients for determining deformation:

$$
\begin{aligned}
A_1 = {} & \mu R \left[0.417c - 0.139g - 6.48g\cos^2\alpha - 0.556c\log(R)\right. \\
& \left. + g\left(0.556\cos^2\beta - 0.556\cos(2\beta)\log(R)\right)\sin^2\alpha\right]. \\
B_1 = {} & R^2 \left\{ g(-0.828 + 4.830\mu)\cos\beta\log(R)\sin^2\alpha\sin\beta\cos\Theta + 0.198c\sin\Theta \right. \\
& + \left[0.198g + 0.910c\mu + 1.307g\mu - 4.830g\mu\cos^2\alpha - 1.225c\mu\log(R)\right. \\
& \left.\left. + g\mu\left(-0.397\cos^2\beta - 1.225\cos(2\beta)\log(R)\right)\sin^2\alpha\right]\sin\Theta \right\}. \\
C_1 = {} & R^2 \left\{ g(-0.828 + 4.830\mu)\cos\beta\log(R)\sin^2\alpha\sin\beta\sin\Theta - 0.198c\cos\Theta \right. \\
& + \left[-0.198g - 0.910c\mu - 1.307g\mu + 4.830g\mu\cos^2\alpha + 1.225c\mu\log(R)\right. \\
& \left.\left. + g\mu\left(0.397\cos^2\beta + 1.225\cos(2\beta)\log(R)\right)\sin^2\alpha\right]\cos\Theta \right\}. \\
D_1 = {} & -g\mu R^2\cos\beta\sin(2\alpha).
\end{aligned}
\tag{42}
$$

Appendix 5

Energy transformation calculations

The energy transformation process is depicted in **Fig.1**.

To obtain the elongation $\delta\zeta_i$ after each muscle contraction, we need to calculate the energy, and the total energy takes the following form:

$$
\mathcal{E} = \epsilon^2 \int_0^L \mathrm{d}Z \int_S \left(V_0 + \epsilon V_1 + \epsilon^2 V_2\right) R\,\mathrm{d}R\,\mathrm{d}\Theta + O\left(\epsilon^5\right) \tag{43}
$$

where the integration region S is related to each part of the cylinder with a different shear modulus μ, so the model must be divided into 3 different parts for integration.

The final part of the energy conversion per unit volume is then:

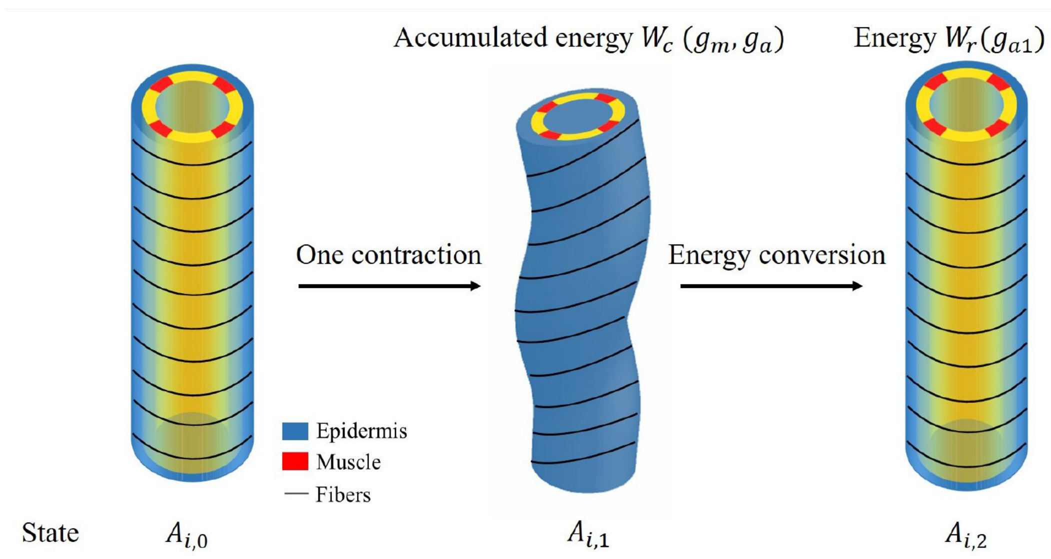
$$
\mathcal{E}_{\oint} = \int_S \left(\epsilon V_1 + \epsilon^2 V_2\right) R\,\mathrm{d}R\,\mathrm{d}\Theta \tag{44}
$$

where V_1 is not the whole first order energy, we only consider the energy induced by activation of acto-myosin g_a and muscles g_m. After obtaining solutions $\mathbf{a}^{(0)}$,$\mathbf{a}^{(1)}$ and deformations from **Eq. (11)**-**(12)**, the accumulated energy during the contractile period W_c that we define is:

$$
\begin{aligned}
W_c &= \int_S \left(\epsilon V_1 + \epsilon^2 V_2\right) R\,\mathrm{d}R\,\mathrm{d}\Theta \\
&= \epsilon\left(-10.70g_a - 7.73g_m\right) + \epsilon^2\left[\left(-3.31 + 26.75g_a\right)g_a + 11.95{g_m}^2\right].
\end{aligned}
\tag{45}
$$

When the muscles are relaxed and only acto-myosin is activated, the total increase of volumetric energy W_r is then:

$$
\begin{aligned}
W_r &= \int_S \left(\epsilon V_1 + \epsilon^2 V_2\right) R\,\mathrm{d}R\,\mathrm{d}\Theta \\
&= -0.41\epsilon g_{a1} + \epsilon^2\left(-5.19 + 6.96g_{a1}\right)g_{a1}.
\end{aligned}
\tag{46}
$$



Appendix 5–figure 1.

Schematic diagram of energy conversion.

By calculating energy conversion, we obtain $g_{a1} = -0.66$ at the beginning of the late elongation phase, **Fig.7** of the manuscript shows the elongation for each contraction and total elongation varies with time.

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Article and author information

Anna Dai
Laboratoire de Physique de l’Ecole normale supérieure, ENS, Université PSL, CNRS, Sorbonne Université, Université Paris Cité, F-75005 Paris, France
ORCID iD: 0000-0002-5655-8885

Martine Ben Amar
Laboratoire de Physique de l’Ecole normale supérieure, ENS, Université PSL, CNRS, Sorbonne Université, Université Paris Cité, F-75005 Paris, France
For correspondence: martine.benamar@phys.ens.fr

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Reviewer #1 (Public Review):

Summary:

The authors have made a novel and important effort to distinguish and include different sources of active deformations for fitting C elegans embryo development: cyclic muscle contractions and actomyosion circumferential stresses. The combination and synchronisation of both contributions are, according to the model, responsible for different elongation rates, and can induce bending and torsion deformations, which are a priori not expected from purely contractile forces. The model can be applied to other growth processes in initially cylindrical shapes.

Strengths:

The model allows us to fit and deduce specific growth patterns, frequencies, and locations of contractions that yield the observed axial elongation during the 240 min of the studied process.

The deformation gradient is decomposed according to muscle and actomyosin activity, which can be distinguished and quantified. An energy-transferring process allows for the retrieval of the necessary permanent deformations that embryo development requires.

Weaknesses:

Despite the completeness of the model, the explanation of the methodology needs to be improved. Parameters and quantities are not always explained in the main text and are introduced on some occasions in an ordered manner. This makes the comprehension and deduction of methodology difficult. There are some minor comments that are listed below. The most important points are:

-How are the authors sure that there is a torsional deformation? Without tracking the muscle fibers, bending with respect to different angles for different Zs may yield a shape similar to the one in Figure 6E. Furthermore, it is unclear why the model yields torsion deformation. If material points of actomyosin rings do not change in reference configuration, no helicoidal growth should be happening.

-The triple decomposition F=F_e*G_i*G_0 seems to complicate the expressions of growth and requires the use of angles alpha and beta due to the initial deformation G_0. Why not use a simpler decomposition F=F_e*G, where G contains all contributions from actomyosin and muscle contractions in a material frame? This would avoid considering angles alpha and beta.

The section "Energy transformation and Elongation" is unclear. Indeed, stresses need to relax, otherwise, the removal of muscle and actin activity would send the embryo back to its initial state. However, the rationale behind the energy transfer is not explained. Authors seem to impose W_c=W_r, and from this deduce the necessary actin contraction after muscle relaxation. Why should energy be maintained when muscle relaxes? Which mechanism physically imposes this energy transfer? Muscle contraction could indeed induce elongation if traction forces at the opposite side of the contracting muscle relax. In fact, an alternative approach for obtaining stress relaxation and axial elongation would be converting part of the elastic deformation F_e to a permanent deformation F_p.

-Self contact is ignored. This may well be a shape generator and responsible for bending deformations. The convoluted shape of the embryo in the confined space deserves at least commenting on this limitation of the model.

Reviewer #2 (Public Review):

Summary:
During C. elegans development, embryos undergo elongation of their body axis in the absence of cell proliferation or growth. This process relies in an essential way on periodic contractions of two pairs of muscles that extend along the embryo's main axis. How contraction can lead to extension along the same direction is unknown.

To address this question, the authors use a continuum description of a multicomponent elastic solid. The various components are the interior of the animal, the muscles, and the epidermis. The different components form separate compartments and are described as hyperelastic solids with different shear moduli. For simplicity, a cylindrical geometry is adopted. The authors consider first the early elongation phase, which is driven by contraction of the epidermis, and then late elongation, where contraction of the muscles injects elastic energy into the system, which is then released by elongation. The authors get elongation that can be successfully fitted to the elongation dynamics of wild-type worms and two mutant strains.

Strengths:
The work proposes a physical mechanism underlying a puzzling biological phenomenon. The framework developed by the authors could be used to explain phenomena in other organisms and could be exploited in the design of soft robots.

Weaknesses:

1. This reviewer considers that the quality of the writing is poor. Because of this the main result of this work, how elongation is achieved by contraction, remains unclear to me. In the opinion of this reviewer, the work is not accessible to a biologist. This is a real pity because the findings are potentially of great interest to developmental biologists and engineers alike.

2. The authors assume that the embryo is elastic throughout all stages of development. Is this assumption appropriate? In my opinion, the authors need to critically discuss this assumption and provide justification. Would this still be true for the adult? If so could the adult relax back to the state prior to elongation? The embryo should be able to do that, if the contractility of the epidermis were sufficiently reduced, right?

3. The authors impose strains rather than stress. Since they want to understand the final deformation, I find this surprising. Maybe imposing strain or stress is equivalent, but then you should discuss this.

4. Does your mechanism need 4 muscle strands or would 2 be sufficient?

5. It is sometimes hard to understand, whether the authors are talking about the model or the worm.

Author Response

> ***Reviewer 1 (Public Review)***
>
> *Summary: The authors have made a novel and important effort to distinguish and include different sources of active deformations for fitting C elegans embryo development: cyclic muscle contrac- tions and actomyosion circumferential stresses. The combination and synchronisation of both contributions are, according to the model, responsible for different elongation rates, and can in- duce bending and torsion deformations, which are a priori not expected from purely contractile forces. The model can be applied to other growth processes in initially cylindrical shapes.*
>
> *Strengths: The model allows us to fit and deduce specific growth patterns, frequencies, and lo- cations of contractions that yield the observed axial elongation during the 240 min of the studied process.*
>
> *The deformation gradient is decomposed according to muscle and actomyosin activity, which can be distinguished and quantified. An energy-transferring process allows for the retrieval of the nec- essary permanent deformations that embryo development requires.*
>
> *Weaknesses: Despite the completeness of the model, the explanation of the methodology needs to be improved. Parameters and quantities are not always explained in the main text and are intro- duced on some occasions in an ordered manner. This makes the comprehension and deduction of methodology difficult. There are some minor comments that are listed below. The most important points are:*
>
> *How are the authors sure that there is a torsional deformation? Without tracking the muscle fibers, bending with respect to different angles for different Zs may yield a shape similar to the one in Figure 6E. Furthermore, it is unclear why the model yields torsion deformation. If material points of actomyosin rings do not change in reference configuration, no helicoidal growth should be happening.*

Our torsional deformations were obtained computationally, and the results are plotted in Figure 6 according to our formalism. In our approach, the torsional deformation results from the interaction between the vertical muscles and the circumferential actin network: the muscles bend the cylinder and the bending modifies the direction of the actin fibers, as demonstrated in the experiment.

> *-The triple decomposition $F = Fe \cdot Gi \cdot G0$ seems to complicate the expressions of growth and requires the use of angles alpha and beta due to the initial deformation $G0$. Why not use a simpler decomposition $F = Fe \cdot G$, where G contains all contributions from actomyosin and muscle contrac- tions in a material frame? This would avoid considering angles alpha and beta.*

$G0$ represents the active strain during the early elongation stage and Gi during the late elongation stage respectively. Such a decomposition which is not mandatory, allows a better un- derstanding. In addition, due to the late elongation stage, both muscle and actin networks must be considered, and their orientation changes with deformation. Therefore, it is clearer and simpler to express the active strain in terms of alpha and beta angles.

> *The section "Energy transformation and Elongation" is unclear. Indeed, stresses need to relax, oth- erwise, the removal of muscle and actin activity would send the embryo back to its initial state. How- ever, the rationale behind the energy transfer is not explained. Authors seem to impose $Wc = Wr$, and from this deduce the necessary actin contraction after muscle relaxation. Why should energy be maintained when muscle relaxes? Which*

mechanism physically imposes this energy transfer? Muscle contraction could indeed induce elongation if traction forces at the opposite side of the contracting muscle relax. In fact, an alternative approach for obtaining stress relaxation and axial elongation would be converting part of the elastic deformation Fe to a permanent deformation Fp.

In this section, we do assume that all the energy accumulated by the muscle contrac- tions will be converted into the energy necessary for elongation, and as our estimate in the article shows, Wc is indeed greater than Wr, indicating that a significant fraction of Wc is converted into dissipation and friction, but also into the reorganization of the actin cables. Indeed, elongation of the cylinder induces a significant reduction in the experimentally observed and also in the actin cable density. However, this reduction in cable density is not observed experimentally. Thus, elon- gation requires a reorganization of the actin network, which is part of the energy consumption and which explains the existence of a permanent deformation Fp.

Self contact is ignored. This may well be a shape generator and responsible for bending deforma- tions. The convoluted shape of the embryo in the confined space deserves at least commenting on this limitation of the model.

Thank you for your suggestion. We have considered the effect of contact between C. elegans and the eggshell in the energy dissipation section but we also agree that the self-contact of the worm in confinement will be important. Here, we focus mainly on active filaments: actomyosin and muscle, and we restrict ourselves to a cylindrical shell that is far from the embryo.

Reviewer 2 (Public Review)

Summary

During C. elegans development, embryos undergo elongation of their body axis in the absence of cell proliferation or growth. This process relies in an essential way on periodic contractions of two pairs of muscles that extend along the embryo's main axis. How contraction can lead to extension along the same direction is unknown.

To address this question, the authors use a continuum description of a multicomponent elastic solid. The various components are the interior of the animal, the muscles, and the epidermis. The different components form separate compartments and are described as hyperelastic solids with different shear moduli. For simplicity, a cylindrical geometry is adopted. The authors consider first the early elongation phase, which is driven by contraction of the epidermis, and then late elongation, where contraction of the muscles injects elastic energy into the system, which is then released by elongation. The authors get elongation that can be successfully fitted to the elongation dynamics of wild-type worms and two mutant strains.

Strengths

The work proposes a physical mechanism underlying a puzzling biological phenomenon. The framework developed by the authors could be used to explain phenomena in other organisms and could be exploited in the design of soft robots.

Weaknesses

> 1. *This reviewer considers that the quality of the writing is poor. Because of this the main result of this work, how elongation is achieved by contraction, remains unclear to me. In the opinion of this reviewer, the work is not accessible to a biologist. This is a real pity because the findings are potentially of great interest to developmental biologists and engineers alike.*

We regret that, despite a general introduction and a number of figures, the work does not seem accessible to biologists.

> 1. *The authors assume that the embryo is elastic throughout all stages of development. Is this assumption appropriate? In my opinion, the authors need to critically discuss this assumption and provide justification. Would this still be true for the adult? If so could the adult relax back to the state prior to elongation? The embryo should be able to do that, if the contractility of the epidermis were sufficiently reduced, right?*

Soft tissues are elastic, the modeling of soft tissues, even with large deformations, is now well established. The difference between a worm embryo and an adult is first of all the quality of the tissues, their low degree of heterogeneity, the weakness of the muscles and the absence of bones. As for the question of complete relaxation of the stresses, the fact that different components are attached to each other limits complete relaxation. We keep our fingerprints and cortical undula- tions, although they originate from an elastic instability that occurs in fetal life. It never disappears.

> *The authors impose strains rather than stress. Since they want to understand the final deformation, I find this surprising. Maybe imposing strain or stress is equivalent, but then you should discuss this.*

Perhaps, the referee has in mind the question of active strain versus active stress and is concerned about the representation of biological forces such as those produced by actomyosin or muscle. In fact, both exist in morphoelasticity and are, of course, related. Usually, the choice is dictated by the simplicity of deriving quantitative results for comparison with experiments.

> 1. *Does your mechanism need 4 muscle strands or would 2 be sufficient?*

First, the 4 muscle strands are consistent with real C. elegans structures, and second, although we assume that two muscles on the same side contract simultaneously, their size and position affect the deformation results. Also, the time period we consider is just before the worm hatches. After that, the worm has to slide on the ground. So efficient muscles are needed.

> 1. *It is sometimes hard to understand, whether the authors are talking about the model or the worm.*

It will be corrected in the new version.